

100 Days of Linux - Day 096

Title: Repeating Tasks with for Loops

Today's Goal

Learn how to repeat commands automatically using **for** loops in Bash scripts.

Why This Matters

For loops help automate repetitive tasks such as processing files, creating folders and running commands multiple times.

Commands of the Day

Syntax:

```
for ITEM in list
do
    commands
done
```

Example:

```
#!/bin/bash
for NAME in Alice Bob Charlie
do
    echo "Hello $NAME"
done
```

Beginner Lesson

A **for** loop repeats the same block of commands for each item in a list. Instead of writing the same command many times, place it inside the loop.

Practice Questions

1. Create a script named `for_loop.sh`.
2. Print the numbers 1 to 5 using a for loop.
3. Print your favourite programming languages from a list.
4. Create folders named `project1`, `project2` and `project3` using a for loop.
5. List the newly created folders using `ls`.

Hints

Use **for**, **do** and **done**. Try **for i in 1 2 3 4 5** or **for i in {1..5}**.

Mini Challenge

Write a script that creates five text files named `file1.txt` through `file5.txt` and then lists them.

Common Mistakes

Forgetting **do** or **done**, incorrect loop syntax or placing commands outside the loop.

Did You Know?

System administrators often use for loops to manage hundreds of files or servers with just a few lines of Bash code.

Pro Tip

Start with small loops while learning, then gradually use them for larger automation tasks.

Answer Key (Instructor Only)

- Q1: `nano for_loop.sh`
Q2: `for i in {1..5}; do echo $i; done`
Q3: `for LANG in Python SQL Linux; do echo $LANG; done`
Q4: `for i in 1 2 3; do mkdir project$i; done`
Q5: `ls`

Homework

Create a script that generates ten folders named `Day1` through `Day10` using a for loop.

Next Day Preview

Tomorrow you'll learn how to repeat tasks using while loops in Bash scripts.